An AI agent forecasts the S&P 500 close

for Fri 20 Nov 2026, scored on the S&P Dow Jones Indices official close

Point forecast, index points

7,637.84

80% interval 7,406 to 7,817

Only the point forecast is scored

Last official close

Thu 12 Nov

7,592.40

Information cutoff

13 Nov, 12:00 UTC

07:00 NY; futures +0.25%

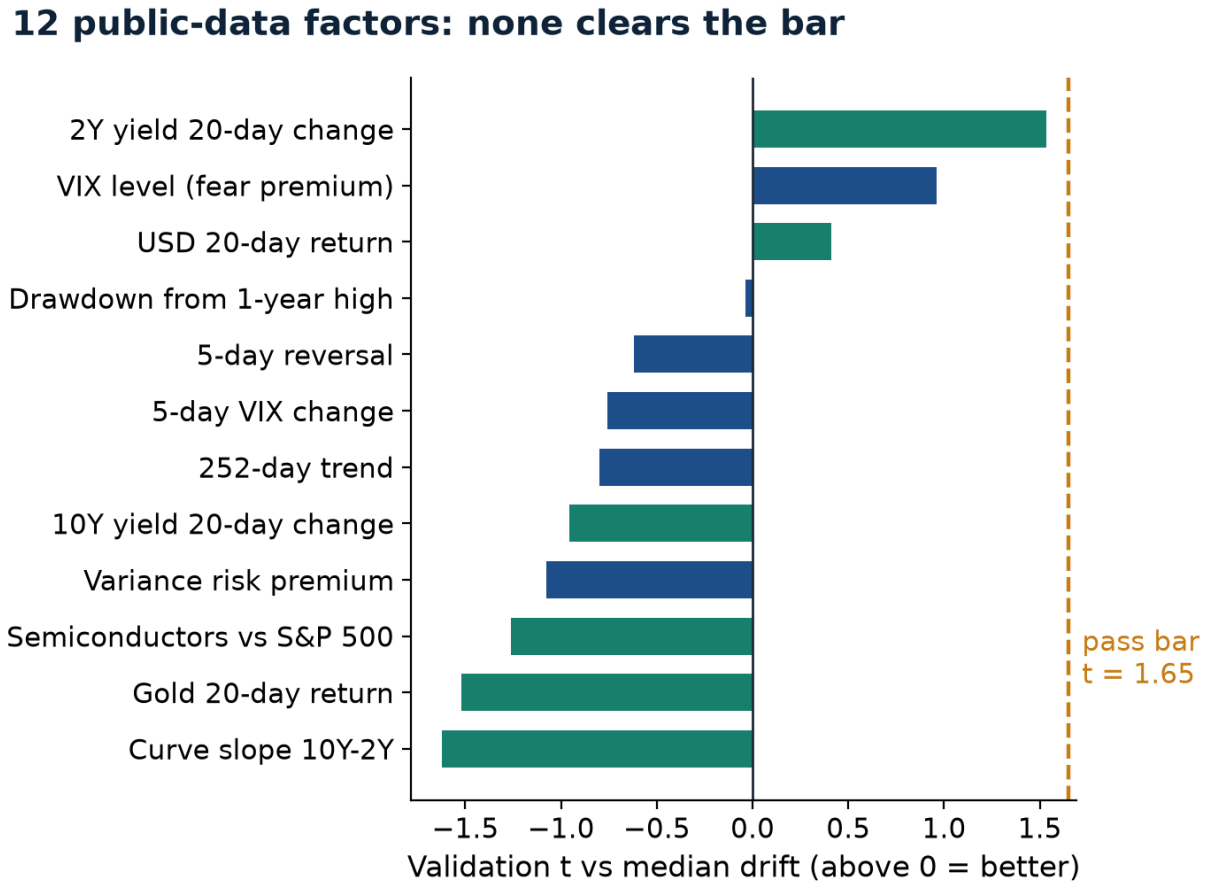
Target close

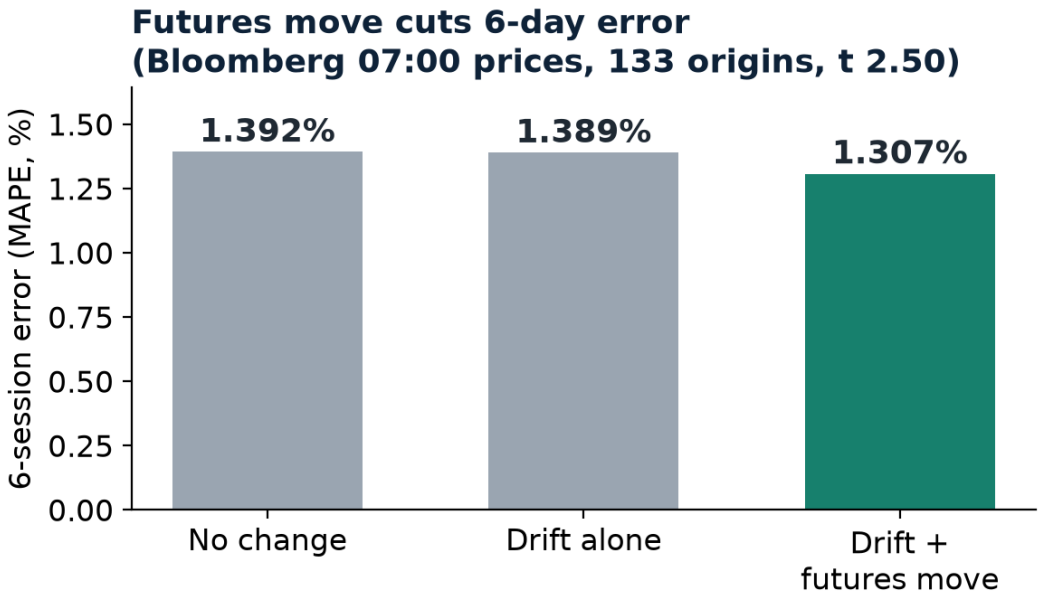
Fri 20 Nov

h = 6 trading sessions

Layla Ahmed | Antonio Herrera Despradel | Jayen Rughani | Mohamed Derbala

Drift is the anchor; futures add the news





Screen: weekly origins 2018-2026, validation to 2021, then 243 untouched test weeks.

0 of 16 factors pass

Best: 2Y yield 20-day change, t 1.53: wrong sign; no 2000-12 history, so below the 2.33 bar. Drift vs no change: t 0.53, a weak edge.

Group checks agree: 30 pre-registered tests, 1.72M-combination search.

Forecast = drift + overnight futures move

7,592.40	× exp(	+0.40%	+	+0.20%	) =	7,637.84
last close (Thu 12 Nov)		median 6-session drift		overnight futures move		forecast

Futures: ES 7,616.25 at the 16:00 close to 7,635.50 at the cutoff = +0.25%, less the average overnight drift (+0.06%) = +0.20%. No hand adjustment.

D Median drift

Why: investors are paid to bear equity risk; negative skew puts the median above zero.

Fails: prolonged bear markets (2000-02, 2008).

N Overnight futures move

Why: futures trade nearly 24h and price overnight news; arbitrage ties the cash open to them.

Fails: contract roll, thin overnight trading, reversal at the open.

Sensitivity of the forecast

No drift (unchanged centre)	**7,607**	-31
Median since 2000	**7,637**	-1
Fixed +0.25% per 6 sessions	**7,626**	-11
Futures read 1 hour earlier	**7,632**	-6
No futures nowcast	**7,623**	-15

80% interval: historical errors scaled by today's VIX.

The agent: who did what, and the proof it ran

1 Goal

HUMAN

Group sets the target, cutoff, data boundary and permitted tools.

2 Data

AGENT

Fetches dated raw files; logs URL, time, SHA-256; drops anything after the cutoff.

3 Tools

AGENT

Python: 16-factor screen, median regression, 447 chronological backtests.

4 Decisions

AGENT

Applies pre-registered rules; logs every choice and correction.

5 Checks

AGENT + HUMAN

Second code path, leak test; humans check the S&P DJI close.

6 Approval

HUMAN

Members re-run, read the trace, sign the record. No trading.

```
>> run_model.py --snapshot data/raw_final --cutoff 2026-11-13T12:
cutoff 2026-11-13 12:00:00+00:00 last close 2026-11-12 = 7592.40
RULE WOULD SELECT: none | FACTOR ENTRY: off (frozen)
"forecast_final": 7637.84,
>> independent_check.py --snapshot data/raw_final --tag final
"PASS_ALL": true
```

From agent_log.md

Correction: 3 faulty Yahoo prices replaced after a Yahoo / Cboe / workbook check.

Decision: model frozen after group review; the screen is reported, never acted on.

Demo (90 s): replay of the agent's /check on the final record (agent/demo_runs/); live on slide 6.

What could invalidate it, and what we claim

Rehearsal outcome

3.9%

Cutoff Fri 30 Oct, target Fri 6 Nov: 7,663 vs realised 7,377; no change missed by 4.7%. Reported, not retuned.

Adverse scenario

-4.1%

Chip rout (index -4%, VIX 30): level 7,324. The interval's lower end is -3.0%.

Counterargument

'Use the news': we tested its measurable parts and none beat drift; the rest is priced in futures. Momentum + RSI (ML) was worse than drift over 22 years. The group's 30 pre-registered tests and 1.72M-combination search agree.

Context and watch list

VIX 18.7; 11/10: October CPI +0.4% (0.3% expected)
- failed independent check or leak test
- futures roll or thin overnight trading
- last close not matching S&P DJI
- Nvidia results 17 Nov, FOMC minutes 18 Nov

We claim

A well-centred, reproducible forecast with honest evidence.

Not foresight: no factor beat drift. Humans sign; the agent never trades.

Questions

Live in Q&A: our agent re-checks the forecast

Claude Code plans its own check on the frozen inputs and recommends; we decide.

1 Its own check — about 30 s

/check: the agent writes a plan, picks its tools (record, recompute, check...), reads each result and recommends submit or not.

2 Hidden fault — about 40 s

The page plants one error in a copy of the record: forecast, last close or cutoff. Nobody knows which. Does the agent catch it? Saved tests (preview and final data): 4 of 4 caught.

3 Your question — about 30 s

Ask anything about the forecast. The agent chooses its own tools and what-if values, then answers.

Live vs fixed

Live: its plan, which tools it calls, the what-if values, every number (Python, run now), its words and its verdict.

Fixed: the frozen data, the tools' code, the goal and the brief's rules.

Boundary and backup

No internet; reads files and runs code/demo_tools.py only; cannot write files or change the record. A member signs.

Backup: Replay a saved run (agent/demo_runs/).

```
python code/demo_ui.py
```